

An Analysis of Diffusion model

Diffusion model

$$q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

Diffusion model -- Part 1

$p_\theta(x_{t-1} | x_t) = N(x_{t-1} | \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$
 $\{\displaystyle p_{\theta}(x_{t-1}|x_t)=\mathcal{N}(x_{t-1}|\mu_{\theta}(x_t,t),\Sigma_{\theta}(x_t,t))\}$ The goal now is to learn the parameters θ
 $\{\displaystyle \theta\}$ such that $p_\theta(x_0)$
 $\{\displaystyle p_{\theta}(x_0)\}$ is as close to $q(x_0)$
 $\{\displaystyle q(x_0)\}$ as possible. To do that, we use maximum likelihood estimation with variational inference.

Diffusion model -- Part 2

$\alpha^{-t} = (1 - \beta_1) \cdots (1 - \beta_t) = e^{\sum_{i=1}^t \ln(1 - \beta_i)} \rightarrow e^{-\int_0^t \beta(s) ds}$
 $\{\displaystyle \{\bar{\alpha}\}_t=(1-\beta_{\{1\}}\cdots(1-\beta_{\{t\}})=e^{\sum_{i=1}^t\ln(1-\beta_{\{i\}})}\}$ to $e^{-\int_0^t \beta(s) ds}$

Diffusion model -- Part 3

In rectified flow, by injecting strong priors that intermediate trajectories are straight, it can achieve both theoretical relevance for optimal transport and computational efficiency, as ODEs with straight paths can be simulated precisely without time discretization.

Diffusion model -- Part 4

Denoising Diffusion Implicit Model (DDIM) The original DDPM method for generating images is slow, since the forward diffusion process usually takes $T \sim 1000$
 $\{\displaystyle T\sim 1000\}$ to make the distribution of x_T
 $\{\displaystyle x_T\}$ to appear close to Gaussian. However this means the backward diffusion process also take 1000 steps. Unlike the forward diffusion process, which can skip steps as $x_t | x_0$
 $\{\displaystyle x_t|x_0\}$ is Gaussian for all $t \geq 1$
 $\{\displaystyle t\geq 1\}$, the backward diffusion process does not allow skipping steps. For example, to sample $x_{t-2} | x_{t-1} \sim N(\mu_\theta(x_{t-1}, t-1), \Sigma_\theta(x_{t-1}, t-1))$
 $\{\displaystyle x_{t-2}|x_{t-1}\sim\mathcal{N}(\mu_{\theta}(x_{t-1},t-1),\Sigma_{\theta}(x_{t-1},t-1))\}$ requires the model to first sample x_{t-1}
 $\{\displaystyle x_{t-1}\}$. Attempting to directly sample $x_{t-2} | x_t$
 $\{\displaystyle x_{t-2}|x_t\}$ would require us to marginalize out x_{t-1}
 $\{\displaystyle x_{t-1}\}$, which is generally intractable. DDIM is a method to take any model trained on DDPM loss, and use it to sample with some steps skipped,

sacrificing an adjustable amount of quality. If we generate the Markovian chain case in DDPM to non-Markovian case, DDIM corresponds to the case that the reverse process has variance equals to 0. In other words, the reverse process (and also the forward process) is deterministic. When using fewer sampling steps, DDIM outperforms DDPM. In detail, the DDIM sampling method is as follows. Start with the forward diffusion process $x_t = \alpha^{-t} x_0 + \sigma_t \epsilon$
 $\{\displaystyle x_t=\sqrt{\alpha_t}x_0+\sigma_t\epsilon\}$. Then, during the backward denoising process, given x_t , $\epsilon_\theta(x_t, t)$
 $\{\displaystyle \epsilon_{\theta}(x_t,t)\}$, the original data is estimated as $x_0' = x_t - \sigma_t \epsilon_\theta(x_t, t) \alpha^{-t}$
 $\{\displaystyle x_0'=\frac{x_t-\sigma_t\epsilon_{\theta}(x_t,t)}{\sqrt{\alpha_t}}\}$ then the backward diffusion process can jump to any step $0 \leq s < t$
 $\{\displaystyle 0\leq s<t\}$, and the next denoised sample is $x_s = \alpha^{-s} x_0' + \sigma_s \epsilon_s$
 $\{\displaystyle x_s=\sqrt{\alpha_s}x_0'+\sigma_s\epsilon_s\}$ where ϵ_s
 $\{\displaystyle \epsilon_s\}$ is an arbitrary real number within the range $[0, \sigma_s]$
 $\{\displaystyle [0,\sigma_s]\}$, and $\epsilon_s \sim N(0, I)$
 $\{\displaystyle \epsilon_s\sim\mathcal{N}(0,I)\}$ is a newly sampled Gaussian noise. If all $\sigma_s' = 0$
 $\{\displaystyle \sigma_s'=0\}$, then the backward process becomes deterministic, and this special case of DDIM is also called "DDIM". The original paper noted that when the process is deterministic, samples generated with only 20 steps are already very similar to ones generated with 1000 steps on the high-level. The original paper recommended defining a single "eta value" $\eta \in [0, 1]$
 $\{\displaystyle \eta\in[0,1]\}$, such that $\sigma_s' = \eta \sigma_s$
 $\{\displaystyle \sigma_s'=\eta\sigma_s\}$. When $\eta = 1$
 $\{\displaystyle \eta=1\}$, this is the original DDPM. When $\eta = 0$
 $\{\displaystyle \eta=0\}$, this is the fully deterministic DDIM. For intermediate values, the process interpolates between them. By the equivalence, the DDIM algorithm also applies for score-based diffusion models.

Diffusion model -- Part 5

$d_t \phi(x) = v_t(\phi(x))$ $t \in [0, 1]$, starting from $\phi_0(x) = x$
 $\{\displaystyle \frac{d}{dt}\phi_t(x)=v_t(\phi_t(x))\}$ $\phi_0(x)=x$
 $\{\displaystyle \phi_0(x)=x\}$

Diffusion model -- Part 6

Such a model is usually trained by presenting it with both (x, y)
 $\{\displaystyle (x,y)\}$ and (x, None)
 $\{\displaystyle (x,\text{None})\}$, allowing it to model both $\nabla_x \ln p(x|y)$
 $\{\displaystyle \nabla_x\ln p(x|y)\}$ and $\nabla_x \ln p(x)$
 $\{\displaystyle \nabla_x\ln p(x)\}$

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$\nabla_x \ln p(x)$. Note that for CFG, the diffusion model cannot be merely a generative model of the entire data distribution $\nabla_x \ln p(x)$. It must be a conditional generative model $\nabla_x \ln p(x|y)$. For example, in stable diffusion, the diffusion backbone takes as input both a noisy model x_t , a time t , and a conditioning vector y (such as a vector encoding a text prompt), and produces a noise prediction $\epsilon_\theta(x_t, y, t)$. For denoising models, it corresponds to $\theta(x_t, y, t, \gamma) = \theta(x_t, t) + \gamma(\theta(x_t, y, t) - \theta(x_t, t))$. As sampled by DDIM, the algorithm can be written as

$$\begin{aligned} \epsilon_{\text{uncond}} &\leftarrow \epsilon_\theta(x_t, t) & \epsilon_{\text{cond}} &\leftarrow \epsilon_\theta(x_t, t, c) \\ \text{CFG} &\leftarrow \epsilon_{\text{uncond}} + \gamma(\epsilon_{\text{cond}} - \epsilon_{\text{uncond}}) & x_0 &\leftarrow (x_t - \sigma_t \text{CFG}) / (1 - \sigma_t^2) \\ x_s &\leftarrow (1 - \sigma_s^2)x_0 + \sigma_s^2 - (\sigma_s')^2 \epsilon_{\text{uncond}} + \sigma_s' \epsilon_{\text{CFG}} \end{aligned}$$

A similar technique applies to language model sampling. Also, if the unconditional generation $\epsilon_{\text{uncond}} \leftarrow \epsilon_\theta(x_t, t)$ is replaced by $\epsilon_{\text{neg cond}} \leftarrow \epsilon_\theta(x_t, t, c')$, then it results in negative prompting, which pushes the generation away from c' condition.